GREAT LAKES BIOINFORMATICS CONFERENCE 13-16 May 2024

Simulation to Characterize the Spatial Dynamics of Cross-Membrane Signal Transduction and Data Visualization

Eileen Deng (Student), Dr. Yinghao Wu (Mentor)

ed437@cornell.edu; yinghao.wu@einsteinmed.edu

Abstract

We present a new generic Monte Carlo simulator, SIMCROST, to characterize the spatial dynamics of cross-membrane signal transduction based on a domain-based mesoscale representation of protein structures. These domains may include extracellular domains, transmembrane domains, intracellular signaling domains (e.g., tyrosine kinase domains of receptor tyrosine kinases (RTKs)), membrane-anchoring domains, or others. We tested the SIMCROST simulator on a hypothetical system, using currently available experimental evidence of RTK as a prototype. We created a dataset with filtered features based on large bulk data (millions of records) generated by multiple simulations in the context of protein structure analysis or molecular dynamics; each simulation data contained 100,00 records. Using visualization tools and Python libraries (like Pandas and Matplotlib), we explored and visualized in a model, the relationships among variables and the 2D or 3D environment in which the simulation took place. Furthermore, these developed datasets can serve as supplementary resources for spatial regulation research applied by machine learning algorithms.

Introduction

Cross-membrane signal transduction is an important functional module of cells for adapting to their surrounding environments or responding to specific stimuli. It plays an indispensable role in many physiological or pathological processes. During these processes, the signals are initiated by the binding between membrane receptors and their ligands in the extracellular region. The binding triggers a series of signaling cascades in the cytoplasm and finally leads to specific outcomes, such as changes in transcriptional regulation. A wide spectrum of theoretical or computational methods have been used due to advantages over test conditions that are difficult to attain in vivo experiments. One of their major disadvantages comes from their huge demands for computational resources. Computational modeling can only deal with systems involving a few molecules within a short timescale. Thus, there is an urgent need to develop a method to bridge the gap between various computational models and compensate for the limitations of experimental techniques. Monte Carlo simulations could bridge this gap by capturing the key aspects of a biological process, such as stochastic variability, spatial heterogeneity, and biological and biochemical complexity. In the context of protein structure analysis or molecular dynamics simulations, visualization technology allows us to concisely and visually represent the data obtained.

Procedure and Methodology

We present a new generic Monte Carlo simulator, SIMCROST, to characterize the spatial dynamics of cross-membrane signal transduction based on a domain-based mesoscale representation of protein structures. We tested the SIMCROST simulator on a hypothetical system, using currently available experimental evidence of RTK as a prototype. The simulations record the interactions between four different proteins: the ligand, receptor kinase, scaffold, and effector. A fixed number of each protein (50) is monitored in each simulation and is able to move outside, inside, and on the membrane of the cell. We first run simulations without any changes in conditions and run the simulations again with different conditions to understand the effects the changed conditions have on the interactions. In order to illustrate these interactions, we created a dataset with filtered features based on millions of records generated by multiple simulations in the context of protein structure analysis or molecular dynamics. Using visualization tools and Python libraries (like Pandas and Matplotlib), we explored and visualized in a model, the relationships among variables and the 2D or 3D environment in which the simulation took place.

Data Processing

	index	total interactions	ligand-receptor	receptor-scaffold	scaffold-effector	receptor-receptor
0	100	0	0	0	0	0
1	200	0	0	0	0	0
2	300	0	0	0	0	0
3	400	0	0	0	0	0
4	500	0	0	0	0	0
5	600	0	0	0	0	0
6	700	0	0	0	0	0
7	800	0	0	0	0	0
8	900	0	0	0	0	0
9	1000	0	0	0	0	0
	index	total interactions	ligand-receptor	receptor-scaffold	scaffold-effector	receptor-receptor
99990	9999100	33	25	4	4	0
99991	9999200	33	25	4	4	0
99992	9999300	33	25	4	4	0
99993	9999400	32	24	4	4	0
99994	9999500	32	24	4	4	0
99995	9999600	33	25	4	4	0
99996	9999700	32	24	4	4	0
99997	9999800	33	25	4	4	0
99998	9999900	32	24	4	4	0
99999	10000000	31	23	4	4	0

Multiple files read with glob Libraries with path

```
filenames = glob.glob(path + "/*.csv")
# print('File names:', filenames)
```

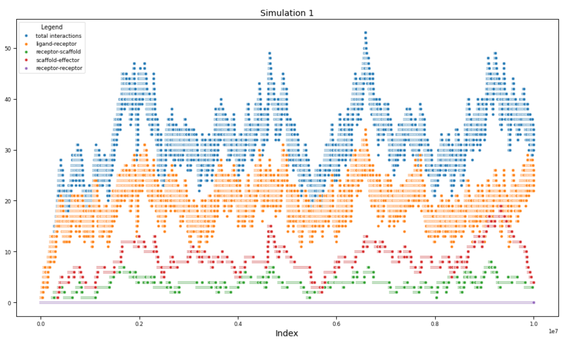
Data visualization by using import matplotlib seaborn libraries.

```
# data visualize for Simulation
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

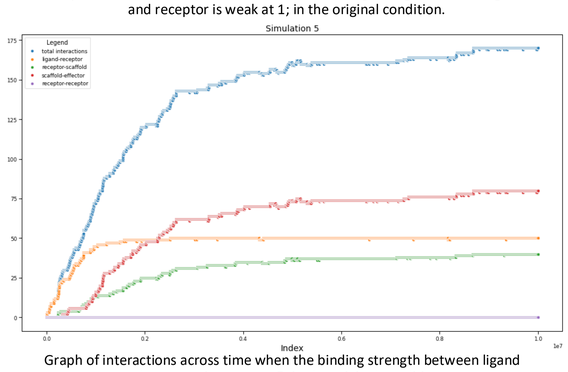
```
plt.figure(figsize=(16, 9))
for i in range(1,len(datapf.columns)):
    sns.scatterplot( data = datapf, x="index",
y= datapf.columns[i], label=datapf.columns[i] )
```

Experiment Setup

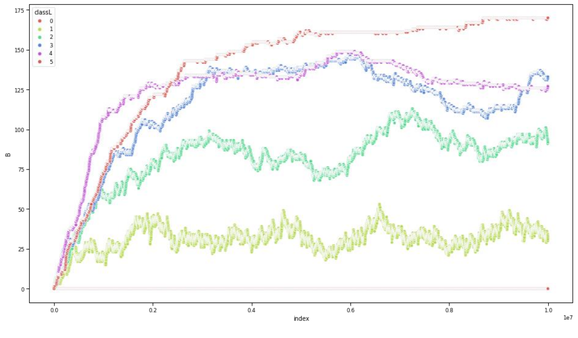
Collect data from simulation
Data Preprocessing (eliminating data samples and define features).
Upload preprocessed data csv into Jupyter local host server cache. Open a new notebook. (Jupyter is a browser-based IDE- integrated development environment).
Import libraries.
Import dataset.
Setup data file path by using Glob Library
Data manipulation and analysis by using Panda Library.
Data visualization and setting target value by Matplotlib plotting library



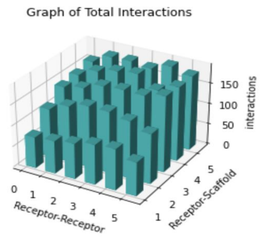
Graph of interactions across time when the binding strength between ligand and receptor is weak at 1; in the original condition.

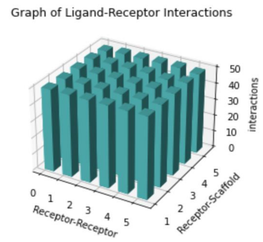


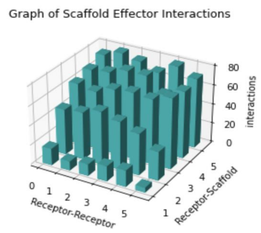
Graph of interactions across time when the binding strength between ligand and receptor is strong at 5; in the original condition.

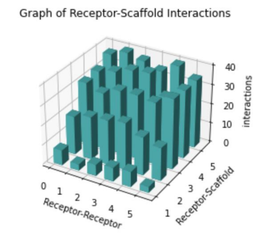


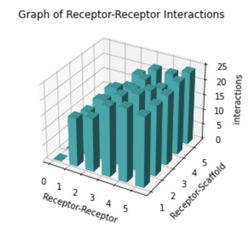
Graph of total interactions that occur across binding strength between the ligand and receptor from 0-5; in the original condition.











3D Histograms of number of receptor-receptor and receptor-scaffold interactions; conditions of increased number of ligands and turning on receptor-receptor interactions.

Conclusion

We explored and visualized in a model. We found that with the changing conditions, the interactions that happened inside the cell were affected. We explored the relationships among variables and the 2D environment in which the simulation took place. We explored the relationships among variables and the 3D environment in which the simulation took place. Furthermore, these developed datasets can serve as supplementary resources for spatial regulation research applied by machine learning algorithms.

Acknowledgement

- **The funding was supported by ISMB and GLBC through a fellowship award, making this presentation of work possible**
- Special thanks to my mentors, Dr. Wu, for his support, encouragement, and guidance and bringing me this opportunity.
- Competing financial interests: The authors declare no competing financial interests.